MEGATHON 2026 · VANGUARD PROBLEM STATEMENT · TEAM 112

Gullak

A student's first money mistake should cost nothing.

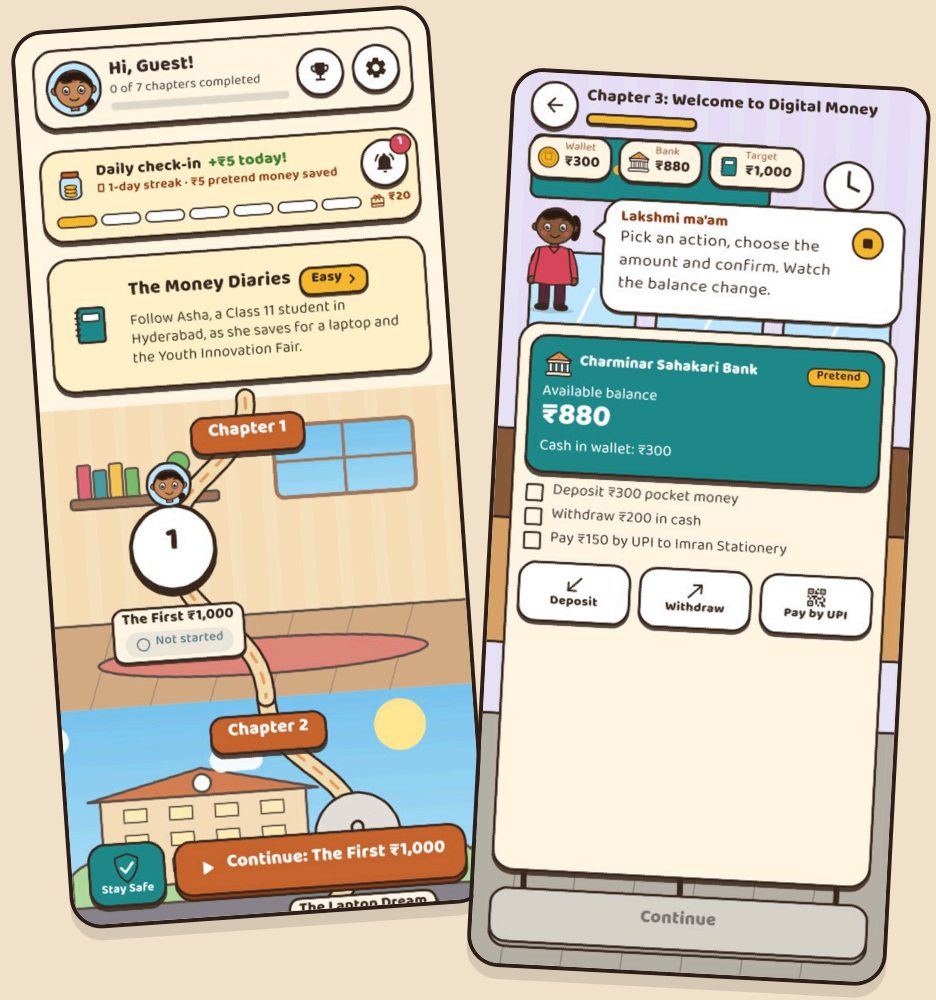
A story game where Grade 11–12 students practise real money decisions with pretend rupees, see the consequences, and try again.

1,500
students in the first rollout, built to scale beyond

₹0
real money and zero personal data at risk

100%
playable offline, in English and Telugu

1 APK
serves every NGO, with a progress dashboard



01 · THE PROBLEM

Real money decisions arrive before any safe practice

Students in underserved Hyderabad communities already handle fares, recharges, fees and UPI requests.
Most have never had a place to make a mistake that costs nothing.

What goes wrong

- A week overspent, with no buffer for a surprise ₹60 cost
- Paying twice because the first UPI payment showed "pending"
- Sharing an OTP with a caller claiming to be the bank
- Buying on credit and paying only the minimum

Why existing options fall short

- **Workshops** need a scheduled session and can't be replayed
- **Booklets and videos** are read-only: no choices, no consequences
- **Finance apps** need data, English, or real money in a real account

What we must respect

- **Low-end, shared phones** with patchy or no data
- **Telugu-first** learners
- **Minors:** no login, no real money, no personal data
- **NGOs** deliver it and need to see progress

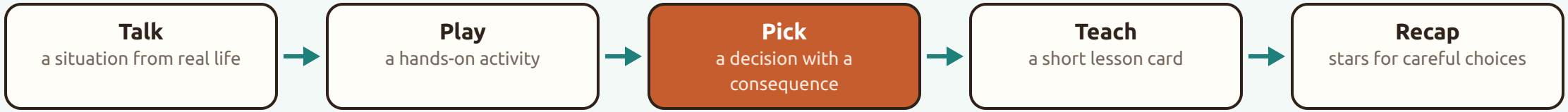
Good content exists. Practice does not.

02 · OUR APPROACH

Make choices. See consequences. Try again.

One continuous story with one wallet. Nothing resets, so every choice matters later, and no choice ever blocks the story.

The learning loop in every chapter



Learn by deciding

A wrong choice shows its cost and offers a recovery path. Stars reward careful decisions, never wealth.

Offline by design

Content, art and storage all live on the phone. The game never needs the internet.

Private by design

No login, no real name. Pretend rupees can't be redeemed, and scam links never open.

NGO-ready

One code links a student to their NGO, with consent. Progress syncs when a connection appears.

Seven connected chapters across all four pillars

Asha, a Class 11 student in Hyderabad, saves for a laptop. Her money and her choices carry from chapter to chapter.

#	CHAPTER	PILLAR	THE STUDENT…
1	**The First ₹1,000**	Plan & budget	builds a week's budget and absorbs a surprise cost
2	**The Laptop Dream**	Save	sets a goal and saves towards it every week
3	**Welcome to Digital Money**	Manage	uses a pretend bank: deposit, withdraw, UPI
4	**The Scholarship Message**	Stay safe	spots the warning signs in fake messages
5	**Buy Now or Wait?**	Stay safe · credit	sees what paying only the minimum really costs
6	**Make Your Money Work**	Investing	compares returns over time against easy access
7	**When Plans Change**	Protect	reads an insurance policy and covers an emergency

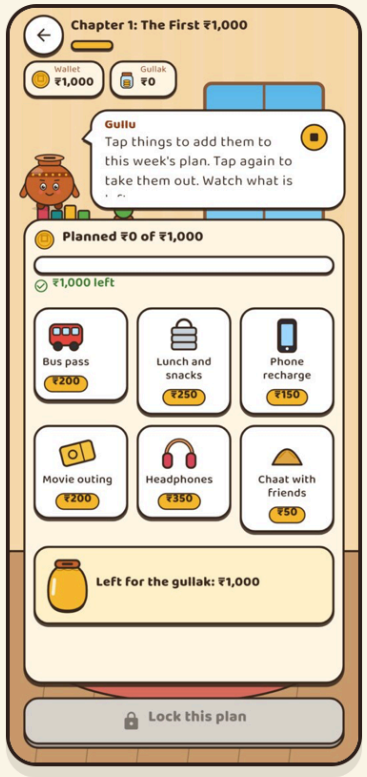
Consequences carry forward

- Money lost to a scam in Chapter 4 shrinks the Chapter 5 budget
- Store credit from Chapter 5 keeps charging interest in Chapter 6
- The saving habit, debts and scam history shape the ending

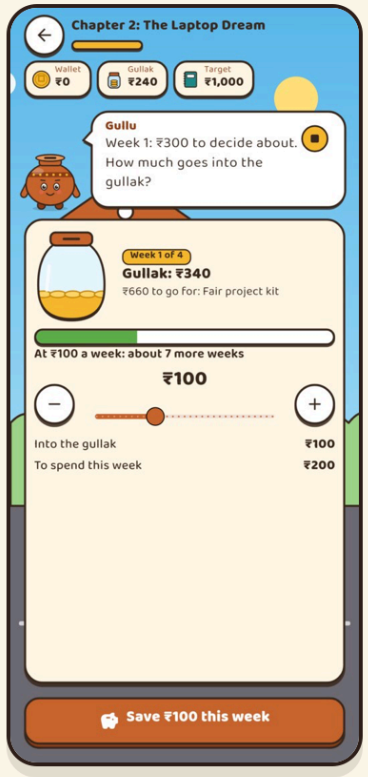
Guardrail respected

Every rate, loan and policy is invented and labelled "pretend" on screen. We educate; we never recommend a product.

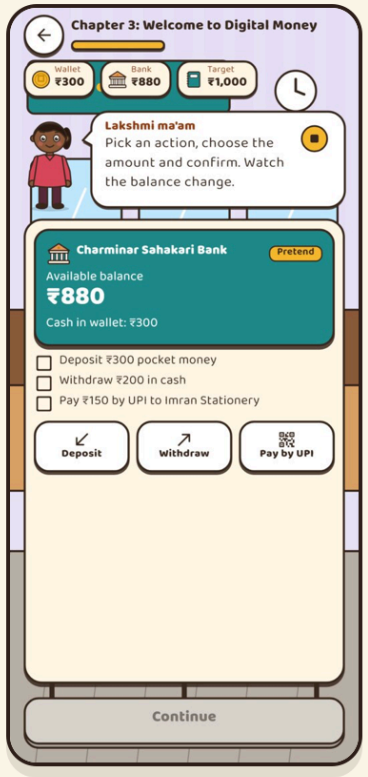
Interactive activities, not multiple-choice quizzes



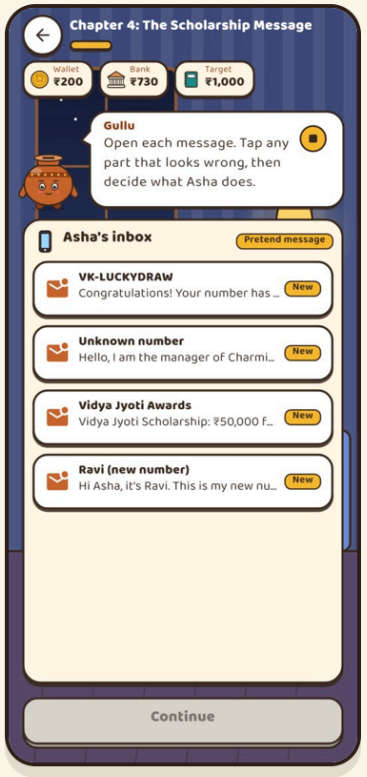
Ch 1 · live budget, 6 items vs ₹1,000



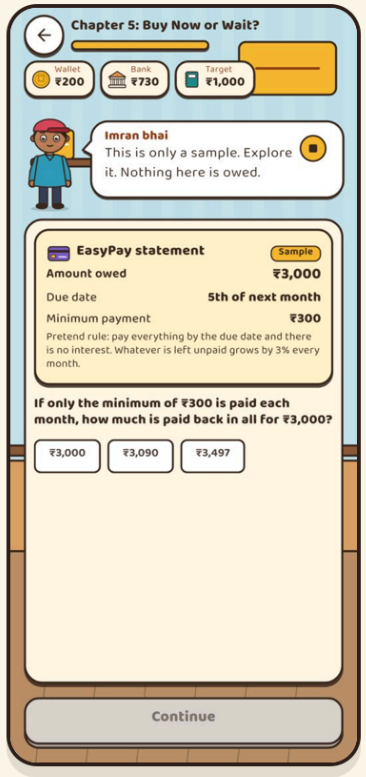
Ch 2 · weekly saving slider fills the jar



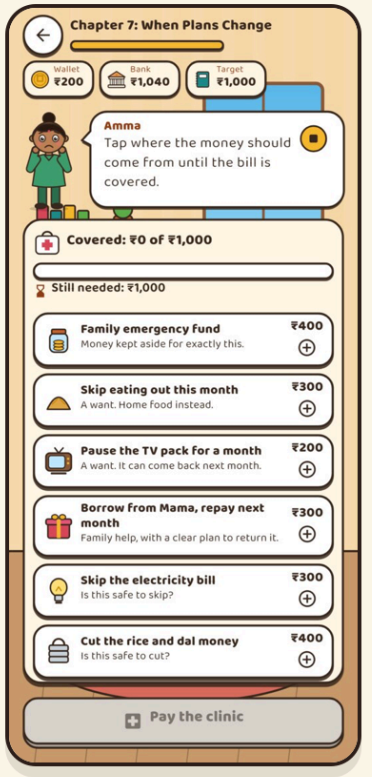
Ch 3 · working pretend bank + UPI



Ch 4 · phishing inbox, tap the red flags



Ch 5 · credit statement, ₹3,000 → ₹3,497



Ch 7 · rebuild the plan for a ₹1,000 bill

Simple & relatable
Bus pass, lunch, recharge, a laptop goal

Language-ready
English and Telugu; a new language needs no code

Persona-led
Profiles by grade, language and reading pace

Game-based
Stars, badges, daily streaks, replays

Intuitive UI
One map, one next button, read-aloud

04 · NGO INTEGRATION

One app for every NGO, with progress they can act on

Coordinators see where students struggle, not who they are.

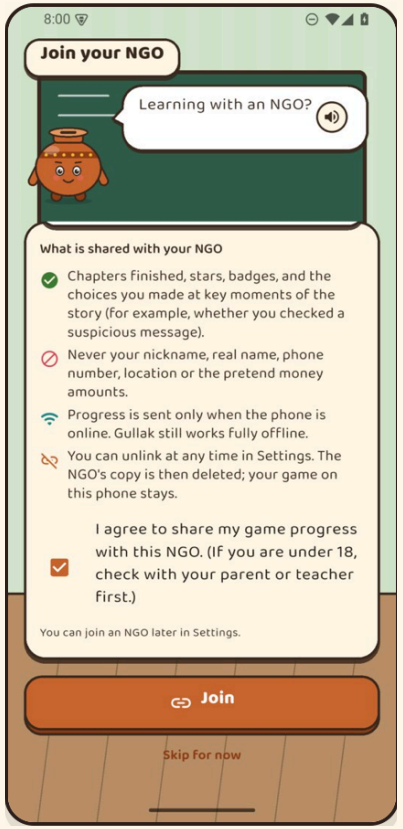
1. **Join with a code**
   The student types their centre's code once and agrees to share. No name, phone number or login.

2. **Play offline**
   Each progress change is queued on the phone in the same step that saves it, so nothing is lost.

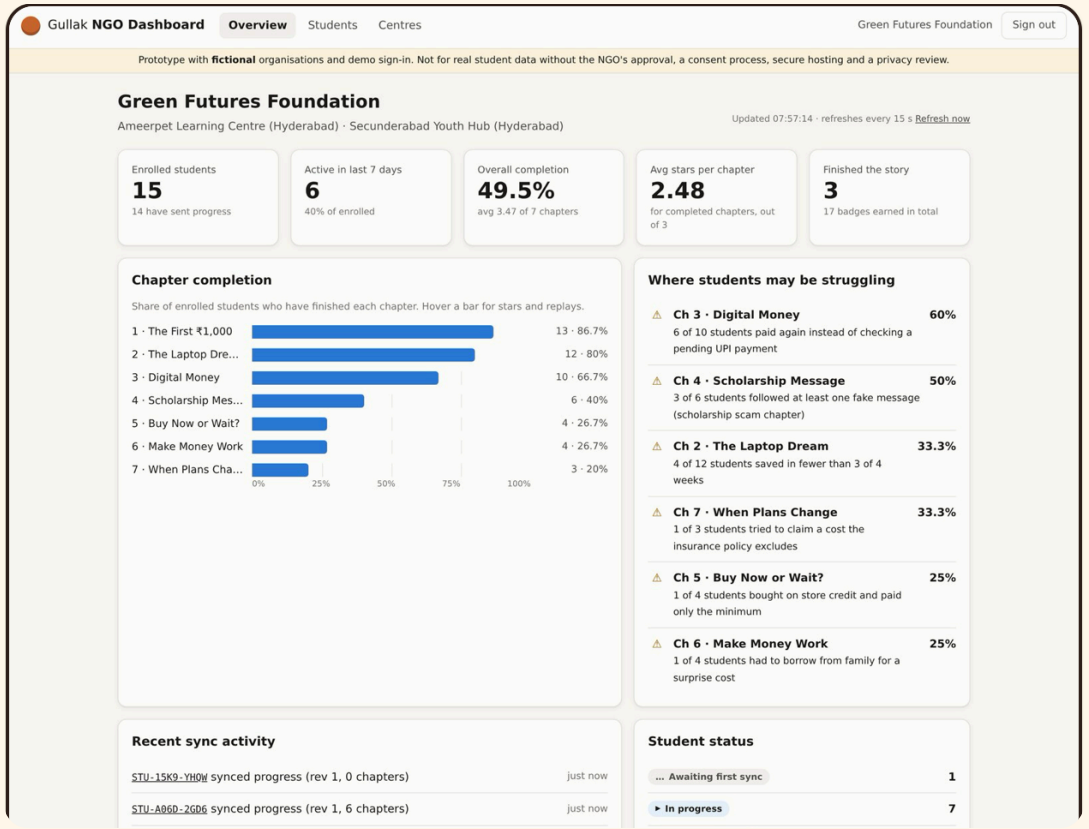
3. **Sync automatically**
   When a connection appears, a tiny summary uploads. Retries can never double-count.

4. **NGO dashboard**
   Chapter completion, students by centre, and the decisions students get wrong most.

Never shared: nickname, grade, pretend balances or the spending diary. Each NGO sees only its own students, and unlinking deletes the server copy.

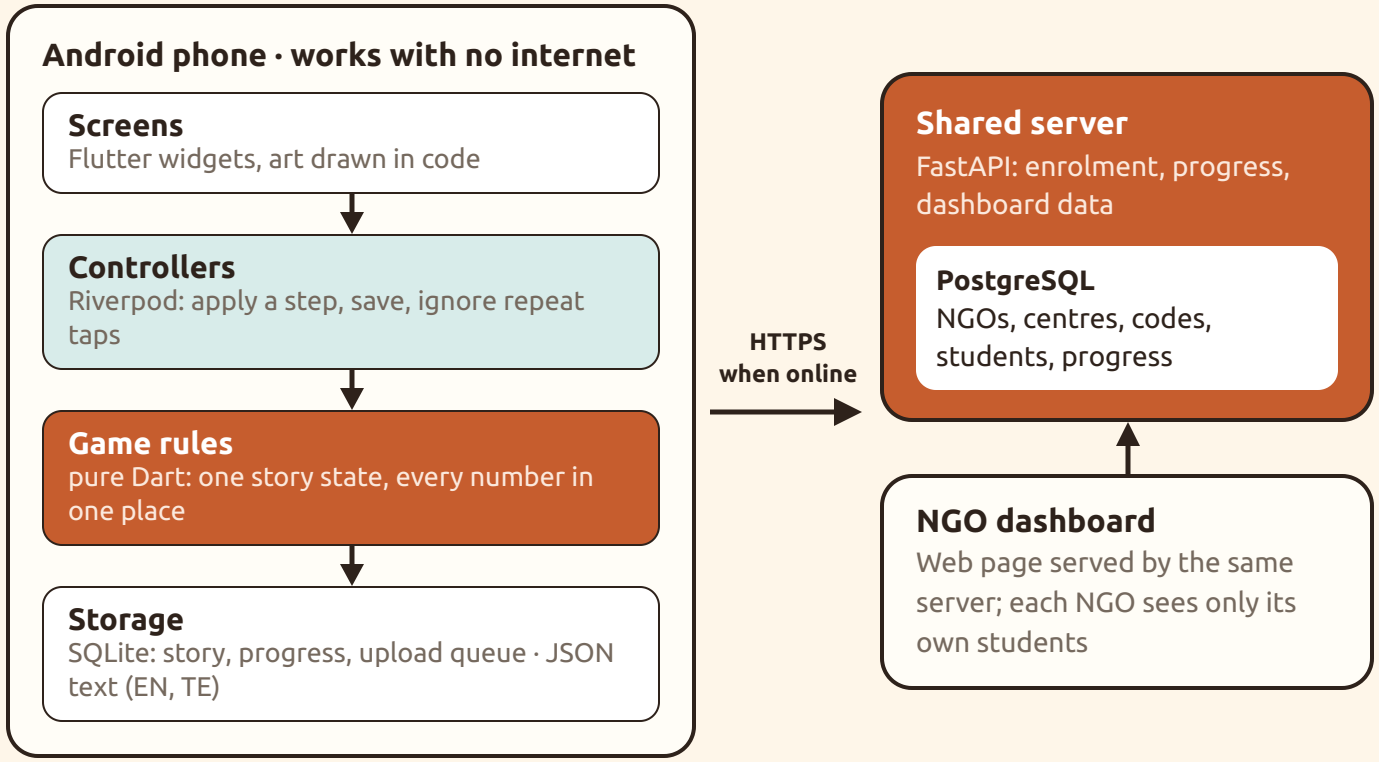


Consent on the phone



The NGO's dashboard (fictional demo data)

Offline-first on the phone, one shared server for every NGO



One source of truth

A single story state holds every balance and choice. Screens never keep their own copy of the money.

Nothing lost offline

A progress change and its queued upload are saved together, then sent when a connection appears.

Safe to retry

Every upload carries a version number. The server ignores repeats, so a poor network never double-counts.

TECH STACK

Flutter · Dart | Riverpod | SQLite | JSON content | Python · FastAPI | PostgreSQL | Render

06 · RESULTS

A working product, verified end to end

7 / 7

chapters playable end to end, in English and Telugu

159

automated app tests passing

32

server tests passing, including isolation between NGOs

0

dead ends on the most careful and the most reckless path

WHAT WE TESTED	WHAT HAPPENED
A reckless student: follows every scam, buys on credit	Ends in debt, **but still finishes the story**
₹3,000 on store credit, paying only the minimum	**₹3,497 repaid over 12 months**, shown to the student
Telugu on a small screen at 130% text size	**No layout breaks** in any chapter
A chapter played in airplane mode, then reconnected	**Synced by itself**; appeared on the right NGO's dashboard
Two fast taps on one decision	Found a double-charge bug; **fixed, with a test**

Fits the phones students have

No image files and no background service. All art is drawn in code.

Honest scope

These are engineering checks on synthetic data. Learning impact with real students is the next thing to measure.

07 · IMPACT AND WHAT COMES NEXT

Ready for a classroom pilot

For the student

- Practises budgeting, saving, banking, scams, credit and insurance before the real thing
- Learns in their own language, on a shared low-end phone, with no data pack
- Mistakes are free and private

For the NGO

- Rollout is one APK and one code per centre
- Sees which topics need a classroom session, by centre
- Holds no names or personal data to protect

Next steps

- **Pilot:** one NGO, two centres, with a before-and-after check
- **Telugu review** of the story text by a fluent speaker
- **More languages:** Hindi and Urdu through the same content files

Levels unlock in order

Finish a level to open the next. Same engine, new story scripts.

Easy Built
budget, save, bank, scams, first credit

→

Medium Planned
credit cards and EMIs, insurance, large expenses

→

Hard Planned
taxes, loans, wedding costs, advanced scams

Practise money decisions safely, before the real ones.